

Milestone 2 Traffic Analysis Report

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Course: CYBR-2102

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Date Submitted: 12/3/25

1. System & Environment Information

Analysis Tool Used: Wireshark

Capture Type: Live Traffic

Capture Duration: 83 seconds

Network Topology Observed: Local workstation capturing internet-bound traffic

File Name of Capture: williamson_capture.pcapng

2. Traffic Capture Overview

Total Packets Captured: 12,157

Protocols Detected: QUIC, TCP, TLS, ICMPv6, DNS, UDP

Highest Volume Protocol: QUIC (61.6%)

Notable Ports Used: 443, 53, various ports

3. Threat / Anomaly Identification Table

Alert ID	Protocol	Description	Severity	Analyst Determination
1	TCP	TCP Analysis Flags errors	Medium	False Positive
2	ICMPv6	Router Advertisements & Neighbor Discovery traffic	Low	False Positive
3	QUIC	High volume QUIC traffic	Medium	True Positive
4	TCP/DNS	DNS over TCP fallback due to blocked or slow DNS responses	Low	True Positive
5	TCP	Malformed packets (likely incomplete capture)	Medium	False Positive

4. Detailed Analysis Notes

Event 1: TCP Analysis Flags

Wireshark detected several TCP retransmissions and reset packets. These usually indicate brief packet loss or congestion on the network. There is no strong evidence of malicious intent.

Event 2: ICMPv6 Router Advertisements

Frequent ICMPv6 Router Advertisements and Neighbor Discovery packets appeared in the capture. These are normal for IPv6-enabled networks as part of internal LAN communication.

Event 3: High Volume QUIC Traffic

QUIC made up over 60% of total traffic. This is expected behavior when using Chrome or Edge as they rely heavily on QUIC for faster encrypted connections. Considered benign.

Event 4: DNS over TCP

A small number of DNS over TCP queries were detected. This usually happens when DNS over UDP fails or is blocked. The traffic appears legitimate.

Event 5: Malformed Packets

Malformed packets were detected (0.4% of traffic). These may appear when captures start mid stream or during encrypted QUIC/TLS traffic. No signs of malicious exploitation.

5. Recommended Defensive Actions

Control Area	Recommendation	Framework Reference
Network Access Control	Limit unnecessary open ports, it ensures the firewall blocks unused services	CIS 4 / NIST AC-4
IDS/IPS Tuning	Add rules to detect repeated retransmissions or abnormal QUIC usage	CIS 7 / NIST SI-4
Patch / Update Action	Ensure all OS and browser updates are applied; QUIC/TLS is regularly updated	CIS 3 / NIST SI-2
Policy / User Training	Educate users on safe browsing, malformed domain names, and phishing	

Wireshark - Protocol Hierarchy Statistics - williamson_capture.pcapng

Protocol	Percent Packets	Packets	Percent Bytes
Frame	100.0	12157	100.0
Ethernet	100.0	12157	1.5
Internet Protocol Version 6	0.4	43	0.0
User Datagram Protocol	0.0	4	0.0
Multicast Domain Name System	0.0	4	0.0
Internet Control Message Protocol v6	0.3	39	0.0
Internet Protocol Version 4	99.6	12108	2.2
User Datagram Protocol	67.9	8249	0.6
QUIC IETF	61.6	7483	60.5
NetBIOS Datagram Service	0.0	2	0.0
SMB (Server Message Block Protocol)	0.0	2	0.0
SMB MailSlot Protocol	0.0	2	0.0
Microsoft Windows Browser Protocol	0.0	2	0.0
Multicast Domain Name System	0.0	4	0.0
Domain Name System	0.5	60	0.1
Data	5.8	700	3.5
Transmission Control Protocol	31.6	3846	0.7
Transport Layer Security	9.1	1109	33.3
Malformed Packet	0.4	48	0.0
Hypertext Transfer Protocol	0.4	44	0.6
Media Type	0.0	1	0.5
eXtensible Markup Language	0.2	30	0.1
Domain Name System	1.6	200	0.2
Internet Group Management Protocol	0.0	5	0.0
Internet Control Message Protocol	0.1	8	0.0
Address Resolution Protocol	0.0	6	0.0

No display filter:CloseCopyProtocolsHelp

williamson_capture.pcapng

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
11984	72.879897	fe80::2e99:24ff:fe9...	ff02::1	ICMPv6	110	Router Adv...
11985	74.787562	192.168.0.10	172.253.63.190	QUIC	1285	Protected
11986	74.787866	192.168.0.10	172.253.63.190	QUIC	274	Protected
11987	74.847578	172.253.63.190	192.168.0.10	QUIC	75	Protected
11988	74.857217	192.168.0.10	172.253.63.190	QUIC	75	Protected
11989	74.861945	172.253.63.190	192.168.0.10	QUIC	118	Protected
11990	74.871999	192.168.0.10	172.253.63.190	QUIC	82	Protected
11991	74.917443	172.253.63.190	192.168.0.10	QUIC	68	Protected
11992	75.342178	fe80::2e99:24ff:fe9...	fe80::b503:48a1:9b6...	ICMPv6	86	Neighbor S...
11993	75.342318	fe80::b503:48a1:9b6...	fe80::2e99:24ff:fe9...	ICMPv6	86	Neighbor A...
11994	75.448945	192.168.0.10	172.64.148.235	TLSv1.3	144	Applicatio...
11995	75.502321	172.64.148.235	192.168.0.10	TCP	56	443 -> 630...
11996	75.528414	172.64.148.235	192.168.0.10	TLSv1.3	132	Applicatio...

> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured on interface 0
> Ethernet II, Src: Intel_2e:cc:3c (20:79:18:2e:cc:3c), Dst: Intel_2e:cc:3c (20:79:18:2e:cc:3c)
> Internet Protocol Version 4, Src: 192.168.0.10, Dst: 172.64.148.235
> Transmission Control Protocol, Src Port: 59251, Dst Port: 443

0000 2c 99 24 90 e5 d0 20 79 10
0010 00 34 44 75 40 00 80 06 30
0020 a0 03 e7 73 00 35 80 d0 e0
0030 ff ff a9 8d 00 00 02 04 00
0040 04 02

Ready to load or capturePackets: 12157 - Dropped: 0

1

Search

3:56 PM

12/3/2025



Conversation Settings

- ☐ Name resolution
- ☐ Absolute start time
- ☐ Limit to display filter

Copy

Follow Stream...

Graph...

Protocol

- ☐ JXTA
- ☐ LTP
- ☐ MPTCP
- ☐ NCP
- ☐ openSAFETY
- ☐ RSVP
- ☐ SCTP
- ☐ SLL
- ☒ TCP
- ☐ Token-Ring
- ☒ UDP
- ☐ USB
- ☐ ZigBee

Filter list for specific type

Ethernet · 12

IPv4 · 81

IPv6 · 4

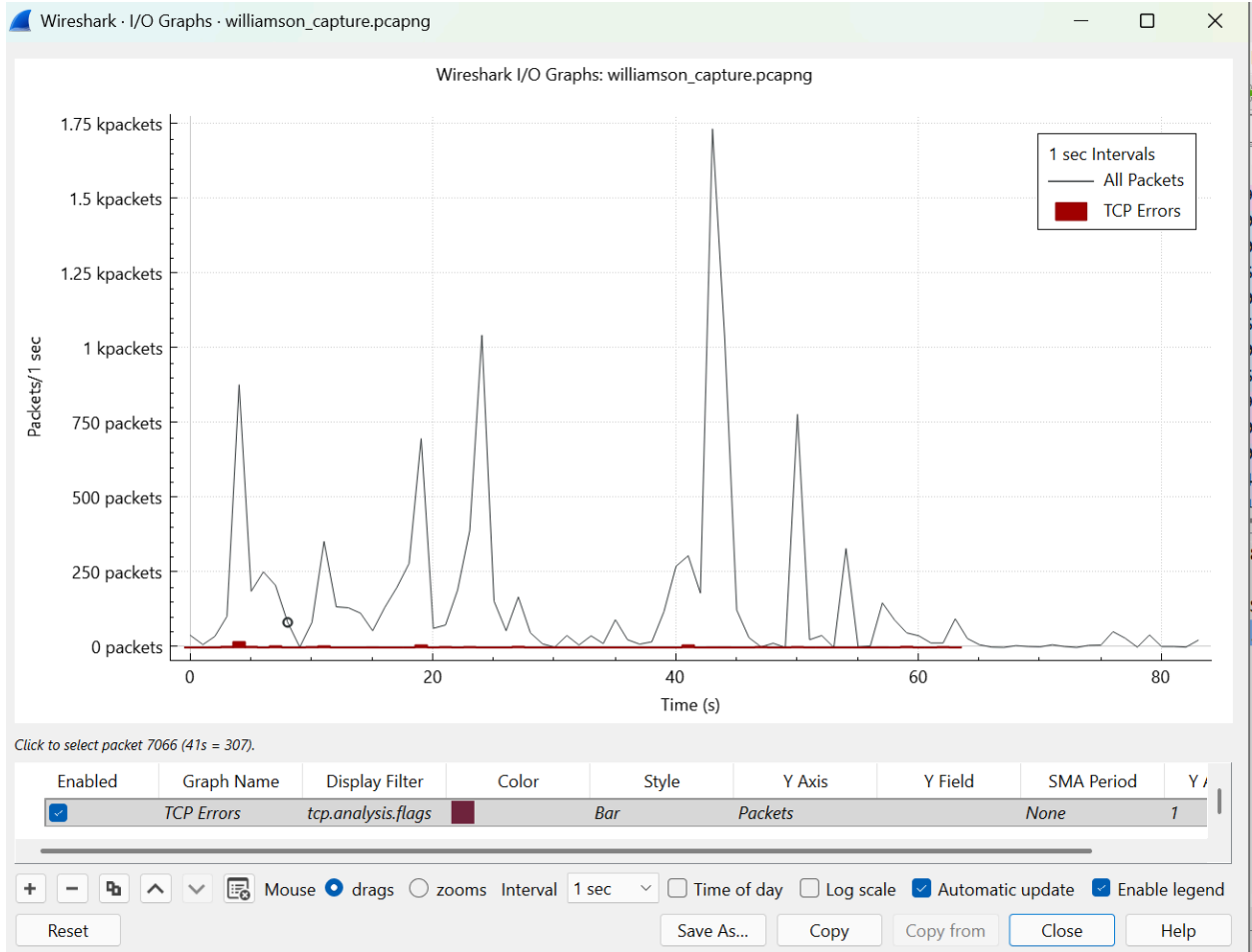
TCP · 212

UDP · 56

Address A	Port A	Address B	Port B	Packets	Bytes
50.16.61.80	443	192.168.0.10	64554	18	1 kB
52.108.8.12	443	192.168.0.10	51563	31	12 kB
52.108.216.29	443	192.168.0.10	54101	35	3 kB
174.129.191.240	443	192.168.0.10	65298	4	252 bytes
192.168.0.1	45753	192.168.0.10	2869	3	228 bytes
192.168.0.1	45754	192.168.0.10	2869	3	228 bytes
192.168.0.1	45755	192.168.0.10	2869	4	304 bytes
192.168.0.1	45756	192.168.0.10	2869	4	304 bytes
192.168.0.10	55761	3.162.112.19	443	23	8 kB
192.168.0.10	65297	3.162.125.82	443	3	164 bytes
192.168.0.10	63135	3.167.69.120	443	21	6 kB
192.168.0.10	64957	3.167.69.120	443	21	11 kB
192.168.0.10	52936	3.170.3.124	443	21	8 kB
192.168.0.10	56862	3.170.3.124	443	18	7 kB
192.168.0.10	56774	4.150.155.223	443	30	16 kB
192.168.0.10	51916	13.107.42.254	443	34	11 kB
192.168.0.10	65102	13.107.213.51	443	28	10 kB
192.168.0.10	54721	13.107.246.51	443	29	14 kB
192.168.0.10	58408	13.107.246.51	443	62	72 kB
192.168.0.10	54078	13.107.246.254	443	21	11 kB
192.168.0.10	55070	20.135.4.169	443	20	16 kB
192.168.0.10	59763	23.53.35.141	443	14	4 kB
192.168.0.10	61600	23.53.35.141	443	15	4 kB
192.168.0.10	59111	23.55.176.89	443	33	7 kB
192.168.0.10	62793	23.55.176.89	443	22	6 kB
192.168.0.10	51010	23.222.201.252	443	22	11 kB
192.168.0.10	59036	23.222.201.252	443	22	14 kB
192.168.0.10	63911	23.222.201.252	443	22	14 kB
192.168.0.10	52408	24.48.160.2	53	10	857 bytes

Close

Help



Wireshark · Capture File Properties · williamson_capture.pcapng

Details

File

Name: C:\Users\Bryan\Documents\williamson_capture.pcapng
Length: 11 MB
Hash (SHA256): 9543979d11ef0e2185679e7cf652fa543714431572a18a30cf307c3e57a96b65
Hash (SHA1): bae569a0c15490ce7e032cec005fb26858e1c920
Format: Wireshark/... - pcapng
Encapsulation: Ethernet

Time

First packet: 2025-12-03 15:51:12
Last packet: 2025-12-03 15:52:35
Elapsed: 00:01:23

Capture

Hardware: Intel(R) Core(TM) i5-8250U CPU @ 1.60GHz (with SSE4.2)
OS: 64-bit Windows 11 (24H2), build 26100
Application: Dumpcap (Wireshark) 4.4.3 (v4.4.3-0-g66d7a52feb06)

Interfaces

Interface	Dropped packets	Capture filter	Link type	Packet size limit (snaplen)
Wi-Fi	0 (0.0%)	none	Ethernet	262144 bytes

Statistics

Measurement	Captured	Displayed	Marked
Packets	12157	12157 (100.0%)	—
Time span, s	83.164	83.164	—
Average pps	146.2	146.2	—
Average packet size, B	921	921	—
Bytes	11196606	11196606 (100.0%)	0

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williamson_capture.pcapng

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Apply a display filter ... <Ctrl-/>

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11996	75.578414	172.64.148.235	192.168.0.10	TLSv1.3	123	App...

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> Internet Protocol Version 4, Src: 192.168.0.10, Dst: 172.64.148.235
> Transmission Control Protocol, Src Port: 443, Dst Port: 443

Ready to load or capture Packets: 12157 · Dropped: 0 (0.0%) Profile: Default

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